

Food Waste



Reduction Assistant

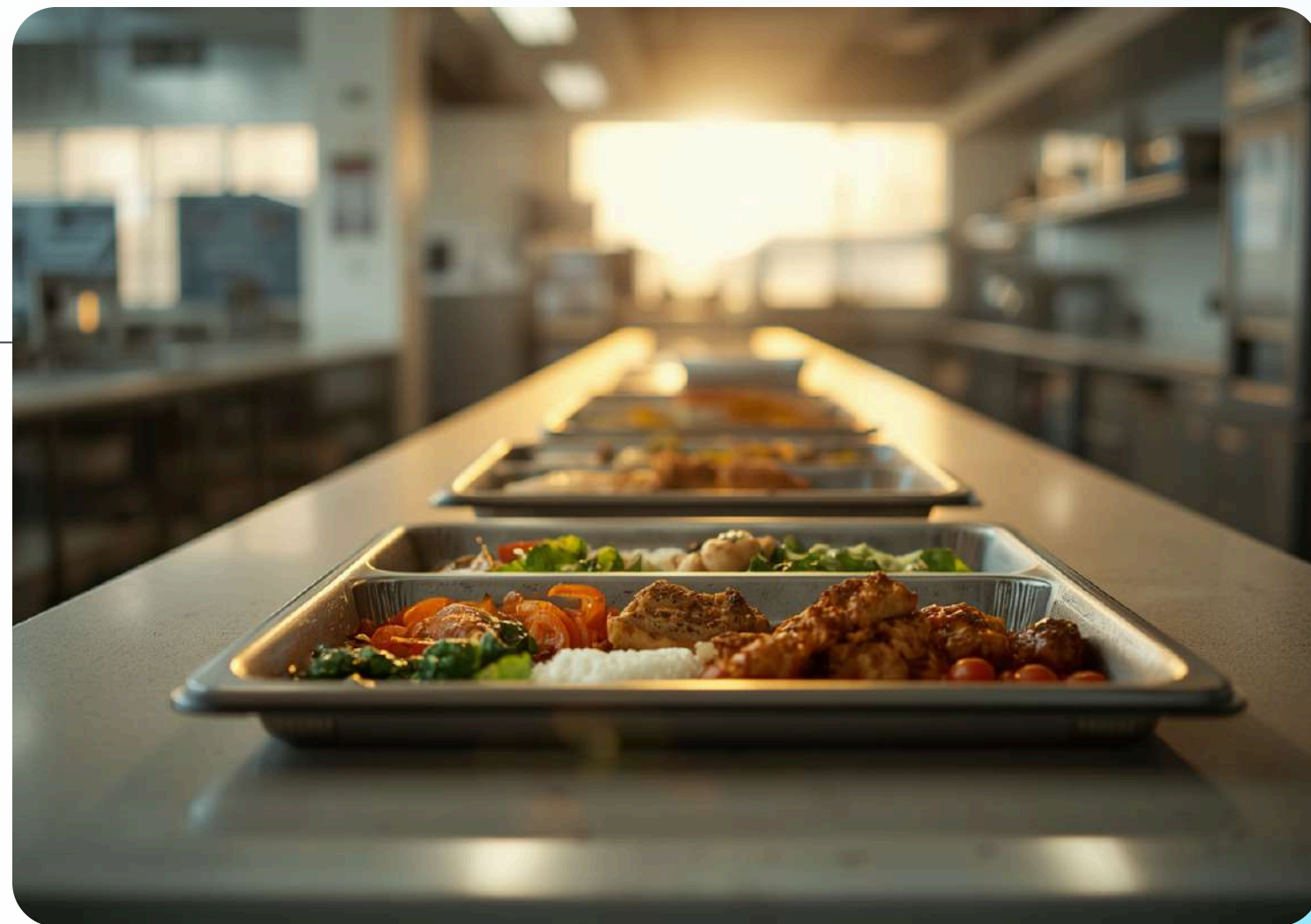


AI for Sustainability Virtual Internship

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Project Overview

The Problem & AI Approach

Food waste is a major issue in educational institutions due to over-preparation of meals. AI-driven systems can analyze consumption patterns, predict demand, and help canteens optimize food preparation decisions.

Analyze consumption patterns

Predict meal demand accurately

Reduce waste & conserve resources

AI helps canteens move from guesswork to data-driven food planning.



SDG 12 – Responsible Consumption

Our project directly supports SDG 12 by reducing food waste and promoting sustainable resource use.

- Reduces food waste
- Conserves resources
- Promotes sustainability



SDG Alignment

Problem Statement



Core Challenge

How might we use AI to analyze canteen food consumption patterns so that food waste can be reduced and resources can be used more sustainably?

30–40%

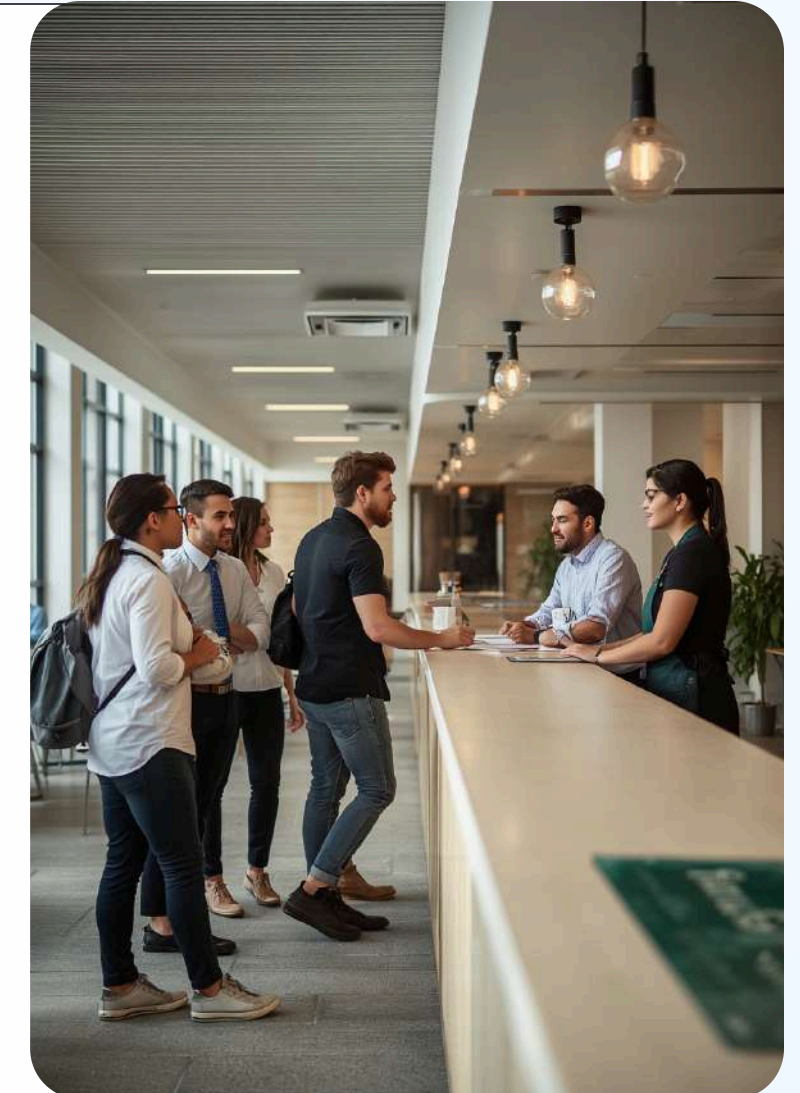
of food prepared in college canteens goes to waste due to over-preparation

No demand prediction leads to excess costs and environmental impact



Target Users

- **Canteen Managers & Kitchen Staff**
- **College Administration & Sustainability Teams**
- **Students & Campus Community**



Key Challenges Faced

- Stakeholders face overproduction, resource wastage, and rising operational costs due to inaccurate meal demand forecasting.

IMPACT

- Overproduction leads to up to 40% excess food prepared daily in college canteens, driving up costs and waste.

Why Food Waste Happens

Without AI-driven forecasting, canteens consistently over-prepare food — leading to waste, higher costs, and environmental harm.

Key Issues

- No demand prediction
- Excess food preparation
- Higher operational costs
- Environmental impact



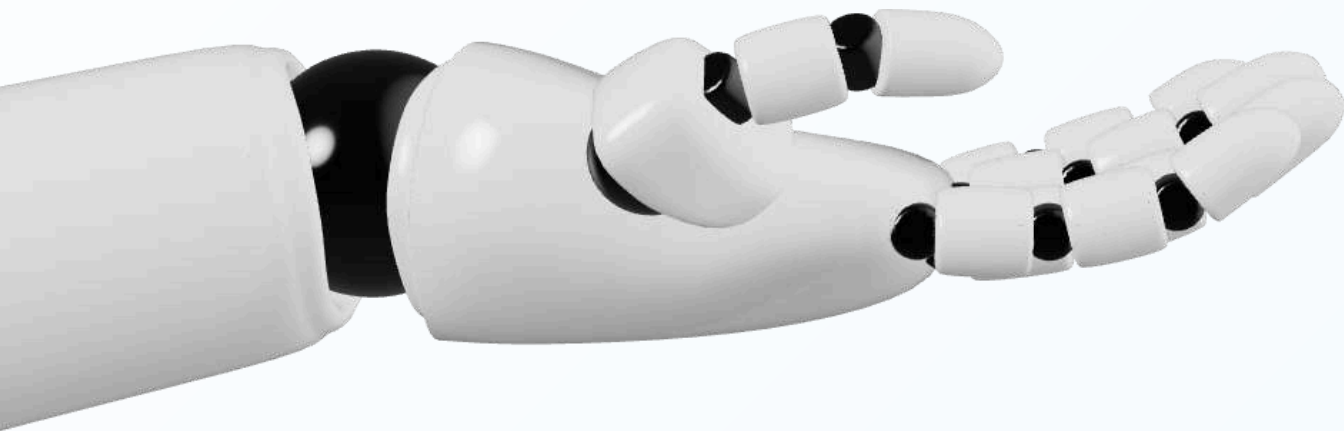
Current Challenges



Dashboard



AI Solution Overview



Why AI? Key Functions & Capabilities

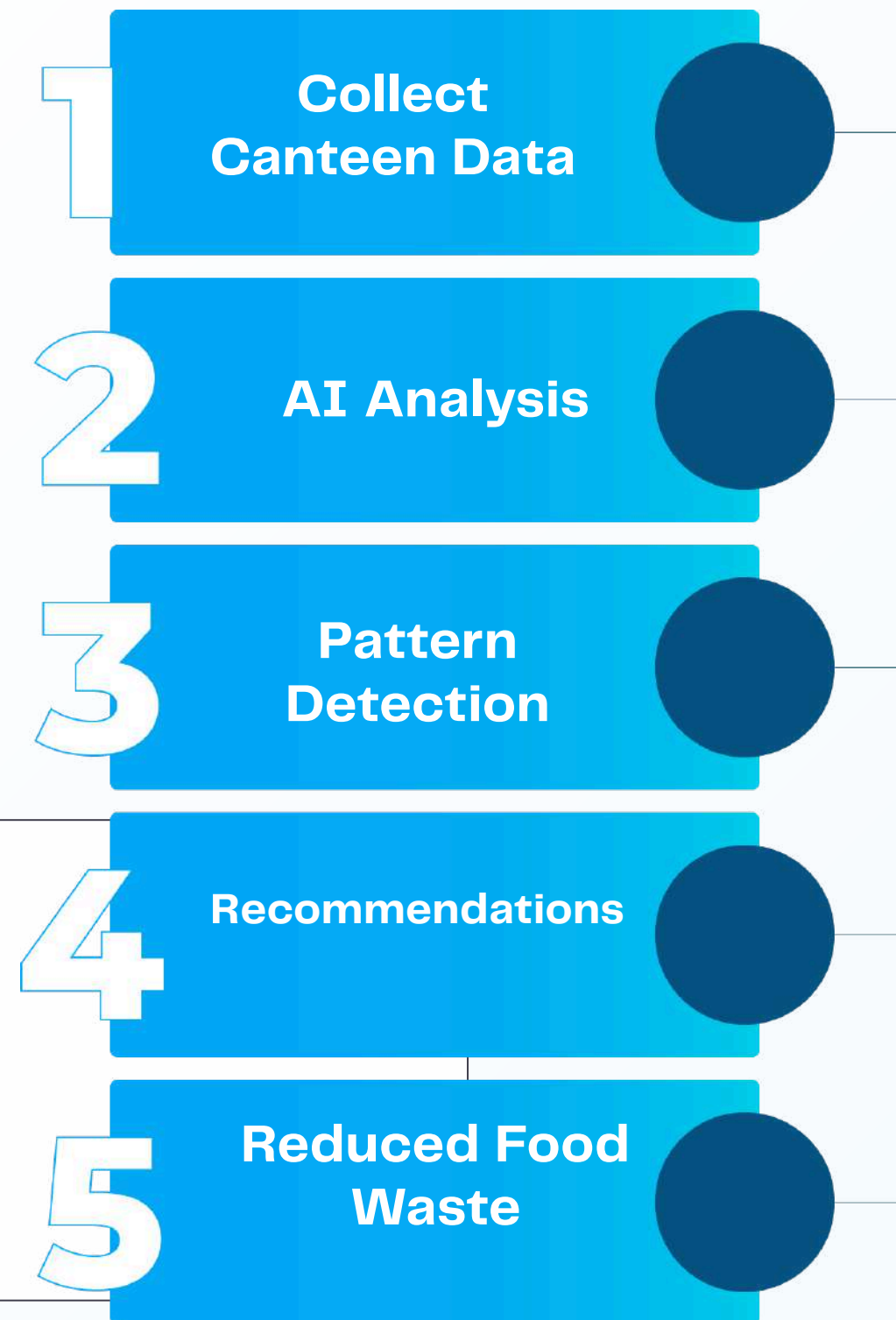
- AI detects hidden patterns in canteen consumption data and predicts future demand accurately – enabling smarter food preparation decisions that minimize waste and optimize resources sustainably.

Pattern Detection → identifies consumption trends from historical data

Demand Prediction → forecasts daily meal requirements accurately

Recommendation Generation → suggests optimal preparation quantities





AI Process Flow – Step by Step

- From raw canteen data to actionable food waste reduction recommendations.

- Demand Prediction
- Recommendations
- Reduced Waste

Workflow Diagram

Chatbot Demo



Live Prototype Interaction

- AI assistant guiding canteen staff with real-time waste tracking and reduction recommendations.

Conversation

User: "We prepared 500 meals and served 450."

Bot: "Food waste is 10%. Reduce tomorrow's preparation by 8–10%."

User: "Which day has the highest waste?"

Bot: "Wednesday has the highest average waste."

AI provides instant, actionable insights for smarter food planning.

Ethics & Trust

- Our AI assistant is designed with responsible principles to ensure fair, transparent, and ethical outcomes for all users.
 - Responsible AI builds trust and ensures sustainable, ethical impact.

Responsible AI Considerations

Core AI Principles

- **Fairness – Equal application of recommendations**
- **Transparency – AI explains its suggestions clearly**
- **Ethics – Avoids harmful or biased decisions**
- **Privacy – No personal student data collected**



Expected Impact



Key Impact Areas

20–30% reduction in food waste

Lower operational costs for canteens

Improved meal planning accuracy

Sustainable resource use & SDG 12 support

Quantified Outcomes

Less Waste + Lower Cost + Better Planning = Sustainability.
AI-driven decisions transform canteen operations into models of responsible resource use.



- By reducing over-preparation, canteens save costs, minimize environmental impact, and contribute directly to responsible consumption goals.

Conclusion & Future

"AI analyzes consumption, predicts demand, reduces waste, and supports a sustainable future for college canteens."